

Solution Requirements

Date	04 July 2026
Team ID	XXXXXX
Project Name	Smart-Lender-Loan-Eligibility-Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow users to enter applicant details such as gender, income, loan amount, education, credit history, etc.	High
2	Authorization levels	The system shall validate all mandatory fields before prediction.	High
3	External interfaces	The system shall predict loan eligibility using the trained machine learning model.	High

4	Transactions processing	The system shall display whether the loan is Approved or Rejected.	High
5	Reporting	The application shall load the trained model (model.pkl) during startup.	Medium
6	Businessrules, etc.	The system shall preprocess user input before passing it to the ML model.	High
7	Compliance to laws or regulations	The system shall display appropriate error messages for invalid inputs.	Medium
8	<i>Other</i>	The web application shall provide an easy-to-use interface on different devices.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should be generated quickly.	Prediction within 2 seconds

2	Scalability	Support multiple users accessing the application.	Supports at least 100 concurrent users without performance degradation.
3	Security & Data Privacy	User data should be processed securely and not permanently stored.	Secure input handling and data privacy
4	Reliability & Availability	The application should provide consistent predictions.	99% successful prediction rate during testing; no application crashes.
5	Usability & Accessibility	The interface should be simple and easy to understand.	Users can complete a prediction in 3 simple steps with a clear interface.
6	<i>Other</i>	Future support for cloud deployment and API integration.	Easily deployable and extendable